

Power & Safety Notes / Operator Guide

This document focuses on **safe handling and operation** of the Vest Tracker system. It is written to reduce risk and prevent avoidable damage. If you need engineering level power details (rails, wiring, pin maps), see Appendix.pdf.

PS1. Battery type and pack layout

1. Lithium battery pack (commonly 3× 18650 Li-ion cells in series in the reference design; always follow the label on your unit).
2. A 3-cell series pack is roughly 11.1 V nominal (about 12.6 V when fully charged). Capacity and runtime depend on the cells installed.
3. The vest uses two DC-DC buck regulators after the power switch:
4. 5 V rail to ESP32 VIN (main controller power).
5. 3.3 V rail to LoRa radio, GPS module, and rain sensor (sensitive peripherals).
6. Do not power LoRa or GPS from the ESP32 onboard 3V3 pin in the reference build. They need the dedicated regulated 3.3 V rail with proper decoupling capacitors

PS2. Charging instructions (general)

1. Use the approved charging method supplied with your system.
2. Charge on a non-flammable surface (tile, metal tray, concrete) and do not charge unattended, stay nearby.
3. When the charger indicator shows full (often a green LED on hobby chargers), disconnect per the charger instructions.
4. If anything becomes unusually hot, smells odd, swells, or leaks: stop charging immediately and power off the device.
5. Typical charge time for ~3200 mAh cells is often about 3–5 hours, depending on charger current.

PS3. Common charging mistakes and countermeasures

Mistake	Why it is risky	Countermeasure
Using a random USB cable or improvised charger	Wrong voltage/current can overheat cells or fail to charge safely	Use only the charger supplied with your unit. Verify connector polarity before plugging in
Leaving the pack charging overnight unattended	Thermal runaway may go unnoticed	Charge while you are nearby. Disconnect when the charger shows full

Mistake	Why it is risky	Countermeasure
Charging a hot pack (left in sun/vehicle) or very cold pack	Li-ion charging outside safe temperature range increases failure risk	Wait until the pack is near room temperature before charging
Mixing old and new cells, or cells of unknown origin	Unbalanced pack stresses weak cells	Use matched, labelled cells as a set. Replace the entire pack if one cell is damaged
Continuing to charge a warm or swelling cell	Indicates internal damage	Stop immediately. Do not reuse that cell or pack

PS4. Do NOT (critical safety rules)

1. Do not short-circuit the battery (never allow metal objects to touch both terminals).
2. Do not expose to heavy rain or submerge.
3. Do not overcharge, disconnect once charged using the provided charger guidance.
4. Do not puncture, crush, or drop the battery pack.
5. Do not use damaged cells (dented, swollen, leaking, corroded, or torn wraps).
6. Do not reverse polarity when connecting the pack or charger — confirm +/- before every connection.

PS5. Operating the vest (power-on habits)

1. Charge before use. Best practice is to charge the day before deployment.
2. Switch the vest off when you are not actively tracking leaving it on in a bag or hot vehicle drains the pack and can overheat the enclosure.
3. If the vest reboots repeatedly or shuts down suddenly, treat that as a low-battery warning, power off, recharge, and inspect the holder and switch before the next field use.
4. Avoid opening the enclosure in rain or when the interior is damp power off first, dry thoroughly, then re-energize only when safe.

PS6. Common operating mistakes and countermeasures

Mistake	Why it is risky	Countermeasure
Running until sudden shutdown (deep discharge)	Deep discharge harms Li-ion cells and causes reboot loops	Recharge before deployment. Do not rely on “one more hour” in the field
Leaving vest powered on in a closed bag/vehicle	Heat build-up and continuous drain	Switch off when not tracking. store in a cool, dry place

Mistake	Why it is risky	Countermeasure
Assuming the rain sensor means the unit is waterproof	Sensor only reports moisture at the probe	Avoid heavy rain and submersion. See Hardware.pdf rain sensor section
Opening the enclosure in the field without authorization	ESD, moisture, and wiring damage	Only authorized maintainers should open the unit. Operators use the external switch and charging port only

PS7. Receiver / control tower power

1. The receiver is powered by USB or a regulated 5 V supply (not the vest battery pack).
2. The receiver must stay powered during operations, if it is off, no LoRa packets reach Firebase and the app shows stale data.
3. Use a stable USB adapter (wall adapter or powered hub). A flaky phone charger or loose cable can cause intermittent Wi-Fi uploads that look like “signal lost” in the app.
4. Do not power-cycle the receiver unnecessarily during active tracking, uploads pause until Wi-Fi and Firebase reconnect (V2 firmware queues a limited number of messages, see FirmwareSoftware.pdf).

PS8. Common receiver power mistakes and countermeasures

Mistake	Why it is risky	Countermeasure
Turning off receiver at end of day but expecting live map	Stale last seen, no new coordinates	Leave receiver on whenever vests are in use
Sharing a weak phone charger with the receiver	Intermittent uploads, long gaps in last seen	Use a dedicated stable 5 V USB supply
Unplugging during Wi-Fi outage “to fix it”	Lost queue flush opportunity	Wait for V2 backoff/recovery. Check Serial Monitor before power-cycling

PS9. Operating temperature (guidance)

1. Avoid operating or charging in extreme heat or extreme cold (rough guide: roughly 0–45 °C for typical Li-ion use; follow your cell manufacturer if labelled).
2. If performance suddenly drops in cold weather, battery output may be reduced temporarily, allow the pack to warm slightly before judging runtime.

3. Do not leave the vest or spare cells in a closed vehicle under direct sun.

PSA10. Storage and transport

1. Store in a cool, dry place away from direct sunlight.
2. Keep away from flammable materials.
3. For long-term storage, follow the battery manufacturer's guidance (avoid indefinite full-charge storage unless your battery documentation says otherwise). A partial charge (~40–60 %) is often recommended for months-long storage.
4. Transport: carry cells in an insulated pouch. Keep keys, coins, and tools away from exposed terminals. Never transport a swollen or damaged pack in the same bag as the powered vest.

PSA11. Rain / water detection vs waterproofing

1. The rain sensor reports water presence at the sensor plate, not whether the whole system is waterproof.
2. Student / field enclosures are often splash resistant at best, not dive-rated. Avoid heavy rain and submersion.
3. Condensation inside the enclosure or on the sensor plate can read as wet. Treat 'wet=true' as "moisture detected" and interpret in context (see [TroubleshootingGuide.pdf](#)).
4. Wiping the sensor dry is safe when the unit is powered off. Avoid forcing water into the enclosure seams.

PSA12. Signs something is wrong (stop / power off)

If you observe any of the following, stop using the device and follow the emergency steps below:

1. Burning smell, smoke, or unusual heat from the vest, battery, or charger.
2. Battery swelling, hissing, or leaking electrolyte.
3. Repeated reboot loops on the vest (can indicate low battery or internal power instability).
4. Corrosion, torn cell wrap, or dented cells in the holder.

PSA13. Emergency response (numbered steps)

1. Power off the vest using the external switch (and unplug the receiver USB if the receiver enclosure is hot or smoking).
2. Move people away from the immediate area if you see smoke or smell burning.
3. Do not use water on a lithium fire. Use an appropriate Class D / lithium-rated extinguisher if trained and available. Otherwise evacuate and call emergency services.
4. When it is safe to touch, disconnect the battery pack from the circuit (maintainer task if the enclosure is sealed).
5. Ventilate the area. Isolate the pack on a non-flammable surface outdoors if possible.

6. Do not reuse swollen, leaking, or overheated cells, dispose per local regulations.
7. Report the incident to your maintainer with approximate time, what was connected (charger vs in-use), and any photos if safe to take.

Quick reference links

Topic	Document
Physical parts and sensor behavior	Hardware.pdf
GPS indoors, stale data, wet alerts	TroubleshootingGuide.pdf
Pin maps, BOM, capacitors	Appendix.pdf